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Report on:

IMAGE CAPTION GENERATOR- *exploring Worlds through Words and Pictures*

IMAGE CAPTION GENERATOR USING DEEP LEARNING

ABSTRACT

The combination of computer vision and natural language processing in Artificial intelligence has sparked a lot of interest in research in recent years, thanks to the advent of deep learning. The context of a photograph is automatically described in English. When a picture is captioned, the computer learns to interpret the visual information of the image using one or more phrases. The ability to analyse the state, properties, and relationship between these objects is required for the meaningful description generation process of high-level picture semantics. Using CNN -LSTM architectural models on the captioning of a graphical image, we hope to detect things and inform people via text messages in this research. To correctly identify the items, the input image is first reduced to grayscale and then processed by a Convolution Neural Network (CNN). The Flickr Dataset was used. The proposed method for blind individuals is intended to be expanded to include persons with vision loss to speech messages to help them reach their full potential and to track their intellect. In this project, we follow a variety of important concepts of image captioning and its standard processes, as this work develops a generative CNN-LSTM model that outperforms human baselines.

CHAPTER 1

INTRODUCTION

Every day, we are bombarded with photos in our surroundings, on social media, and in the news. Only humans are capable of recognizing photos. We humans can recognize photographs without their assigned captions, but machines require images to be taught first. The encoder-decoder architecture of Image Caption Generator models uses input vectors to generate valid and acceptable captions. This paradigm connects the worlds of natural language processing and computer vision. It's a job of recognizing and evaluating the image's context before describing everything in a natural language like English. Our approach is based on two basic models: CNN (Convolutional Neural Network) and LSTM (Long Short-Term Memory). CNN is utilized as an encoder in the derived application to extract features from the snapshot or image, and LSTM is used as a decoder to organize the words and generate captions. Image captioning can help with a variety of things, such as assisting the visionless with text-to-speech through real-time input about the scenario over a camera feed, and increasing social media leisure by restructuring captions for photos in social feeds as well as spoken messages. Assisting children in recognizing chemicals is a step toward learning the language. Captions for every photograph on the internet can result in faster and more accurate authentic photograph exploration and indexing. Image captioning is used in a variety of sectors, including biology, business, the internet, and in applications such as self-driving cars wherein it could describe the scene around the car, and CCTV cameras where the alarms could be raised if any malicious activity is observed. The main purpose of this research article is to gain a basic understanding of deep learning methodologies.

1.1.2 Objectives

1. The project aims to work on one of the ways to context a photograph in simple English sentences using Deep Learning (DL).

2. The need to use CNN and LSTM instead of working with RNN 1.1.3 Scope
Our project extends and is being used in any large-scale business industry and also small-scale business industry.

1.2 Statement Problem In our world, information is considered valuable and some humans face a serious problem regarding visualizing an image. We hence dig into this matter, considering blindness as a major factor, and generate a sentence by allowing users to upload or scan a visual image.

Advantages

- Recommendations for Editing Applications
- Assistance for visually impaired
- Social Media posts
- Self-Driving cars
- Robotics
- Easy to implement and connect to new data sources.

Disadvantages

- Do not make intuitive feature observations on objects or actions in the image
- Nor do they give an end-to-end mature general model to solve this problem

CHAPTER 2

LITERATURE SURVEY

Literature survey is the most important step in the software development process. Before developing the tool, it is necessary to determine the time factor, economy, and company strength. Once these things are satisfied, then the next step is to determine which operating system and language can be used for developing the tool. Once the programmers start building the tool, they need a lot of external support. This support can be obtained from senior programmers, books, or websites. Before building the system, the above considerations are taken into account for developing the proposed system. The major part of the project development sector considers and fully surveys all the required needs for developing the project. For every project, a Literature survey is the most important sector in the software development process. Before developing the tools and the associated designing it is necessary to determine and survey the time factor, resource requirement, manpower, economy, and company strength. To improve and tailor the user experience on its products, photos use image classification. Intra-class variation, occlusion, deformation, size variation, perspective variation, and lighting are all frequent issues in computer vision that are represented by the picture classification problem. Methods that work well for picture classification are likely to work well for other important computer vision tasks like detection, localization, and segmentation as well. Image captioning is a great illustration of this. Given an image, the image captioning challenge is to generate a sentence description of the image. The picture captioning problem is comparable to the image classification problem in that it expects more detail and has a bigger universe of possibilities. Image classification is used as a black box system in modern picture captioning systems, therefore greater image classification leads to better captioned. The image captioning problem is intriguing in and of itself because it brings together two significant AI fields: computer vision and natural language processing. An image captioning system demonstrates that it understands both image semantics and natural language. Once these things are satisfied and fully surveyed, then the next step is to determine about the software specifications in the respective system such as what type of operating system the project would require, and what all the necessary software is needed to proceed with the next step such as developing the tools, and the associated operations. To construct an image sentence, image classification is a key stage in the object recognition and picture analysis process. The final output of the image categorization phase

might be a statement. To date, a variety of image captioning techniques have been presented. Several studies have been carried out in attempt to determine the best image captioning technique. It's difficult to pick one approach as the finest of them all because the results and accuracy are dependent on a variety of circumstances. In order to achieve the most accurate results, traditional approaches have been constantly modified as well as new image captioning techniques invented during the previous few decades. Each caption generator technique has its own set of benefits and drawbacks. The focus of the research today is on combining the desired qualities of various techniques in order to boost efficiency. Many high-level tasks, such as image classification, object detection, and, more recently, semantic segmentation, have recently been proven to obtain outstanding results using convolutional neural networks with many layers. A two-stage technique is frequently used, especially for semantic segmentation. Convolutional networks are trained in this way to offer good local pixel-wise data for the second stage, which is often a more global graphical analysis model. We will use Long short-term memory (LSTM), which is a subset of RNNs, to tackle the problem of Vanishing Gradient. The main goal of LSTM is to solve the problem of Vanishing Gradients. The unique feature of LSTM is that it can keep data values for long periods, allowing it to address the vanishing gradient problem.

When compared to applying RNN, the results revealed that using a mixture of LSTM generated better outcomes. CNNs employ multilayer convolution to accomplish feature engineering and integrate these features internally, unlike traditional image recognition algorithms.

CHAPTER 3

AIM AND SCOPE OF PRESENT INVESTIGATION

3.1 Aim Development of automatically describing an image with more than natural language sentences which leads to faster information transfer.

3.2 Scope The application of image captioning in deep learning and securing to become common computing power is one of the main factors that led to techniques for analysis of new and diverse digital data spreading information and response toward users for quick understanding of information with just an image.

3.3 System Requirements To be used efficiently, all computer software needs certain hardware components or other software resourced to be present on a computer. These prerequisites are known as computer system requirements and are often used as guidelines as opposed to absolute rules. Most software defines two sets of system requirements. Minimum and recommended. With the increasing demand for higher processing power and resources in newer versions of software, system requirements tend to increase over time. Industry analysts suggest that this trend plays a bigger part in driving upgrades to existing computer systems than technological advancements.

CHAPTER 4

PROJECT IMPLEMENTATION, ALGORITHMS, AND METHODOLOGY

4.1 Introduction This project is loaded with CNN and LSTM which act as the platform to generate the sentences from a simple image.

This can be worked on all applications.

4.2 Hardware Requirements :

- System: i3 Processor
- Hard Disk: 500 GB.
- Monitor: 15''LED
- Input Devices: Keyboard, Mouse
- Ram: 4GB

4.3 Software Requirements

- Platform: Google Colab
- Coding Language: Python

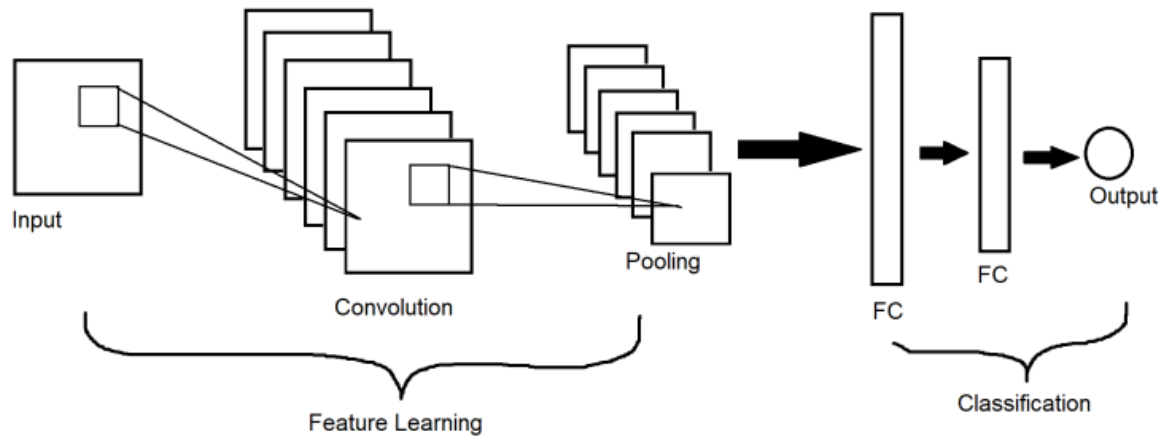
4.4 Working Explanation

1. A user uploads an image that they want to generate a caption for.
 2. A gray-scale image is processed through CNN to identify the objects.
 3. A gray-scale image is processed through CNN to identify the objects.
 4. CNN scans images left-right, and top-bottom, and extracts important image features.
 5. By applying various layers like Convolutional, Pooling, Fully Connected, and thus using activation function, we successfully extracted features of every image.
 6. It is then converted to LSTM.
 7. Using the LSTM layer, we try to predict what the next word could be.
 8. Then the application proceeds to generate a sentence describing the image
- Algorithms

- Convolutional Neural Network
- Long Short-Term Memory

Overview on **CNN Convolutional Neural Network (CNN)** is a type of deep learning model for processing data that has a grid pattern, such as images.

- deep-learning CNN models to train and test, each input image will pass through a series of convolution layers with filters (Kernels), Pooling, fully connected layers (FC), and apply Softmax function to classify an object with probabilistic values between 0 and 1.
- CNNs have unique layers called convolutional layers which separate them from RNNs and other neural networks.
- Within a convolutional layer, the input is transformed before being passed to the next layer. A CNN transforms the data by using filters.



CNN explanation

Some advantages of CNN are:

- It works well for both supervised and unsupervised learning.
- Easy to understand and fast to implement.
- It has the highest accuracy among all algorithms that predict images.
- Little dependence on pre-processing, decreasing the need for human effort to develop its functionalities.

Overview of LSTM

LSTM networks are a type of recurrent neural network capable of learning order dependence in sequence prediction problems. This is a behavior required in complex problem domains like machine translation, speech recognition, and more. LSTMs are a complex area of deep learning. This is a behavior required in complex problem domains like machine translation, speech recognition, and more. LSTMs are a complex area of deep learning.

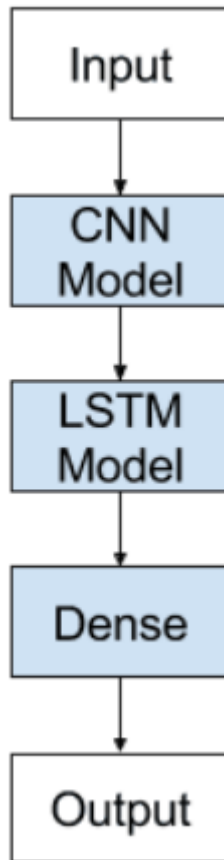
Some advantages of LSTM are:

- Provides us with a large range of parameters such as learning rates, and input and output biases.
- The complexity of updating each weight is reduced to $O(1)$ with LSTMs.

CNN - LSTM Architecture Model

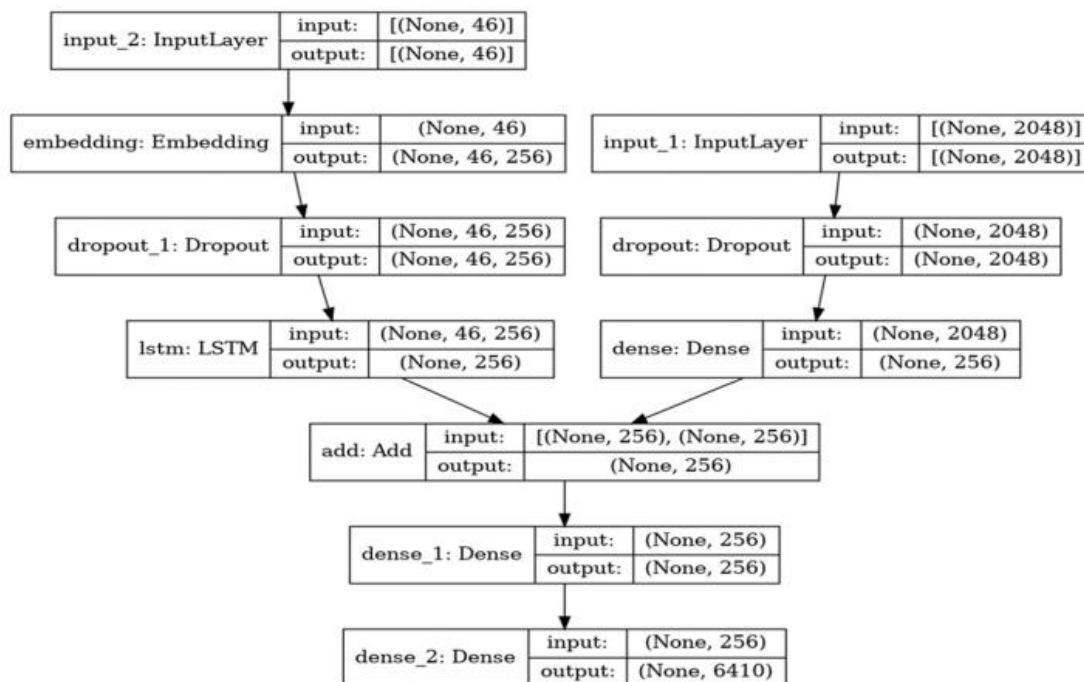
The CNN LSTM architecture involves using Convolutional Neural Network (CNN) layers for feature extraction on input data combined with LSTMs to support sequence prediction. CNN-LSTMs were developed for visual time series prediction problems and the application of generating textual descriptions from a sequence of images (e.g., videos). Specifically, the problem of

- Activity Recognition: Generating a textual description of activity demonstrated in a sequence of images.
 - Image Description: Generating a textual description of a single image.
 - Video Description: Generating a textual description of a sequence of images.
- This architecture was originally referred to as a Long-term Recurrent Convolutional Network (LRCN) model, although we will use the more generic name “CNN LSTM”
- CNN is used for extracting features from the image. We will use the pre-trained model Xception.
 - LSTM will use the information from CNN to help generate a description of the image.



METHODOLOGY

- Import Libraries
- Upload Flickr Dataset from Kaggle. (Data Preprocessing)
- Apply CNN to identify the objects in the image.
- Preprocess and tokenize the captions.
- Use LSTM to predict the next word of the sentence.
- Make a Data Generator
- View Images with captions.



Workflow Diagram

CHAPTER 5

RESULTS AND DISCUSSION

Results: The result of this program is going to be a user being allowed to generate a caption for a visual image using Deep Learning, NLP, and Computer Vision.

CHAPTER 6

SUMMARY AND CONCLUSION

We combined all components of the image caption generation problem in this overview, addressed the model framework proposed in recent years to handle the description task, concentrated on the algorithmic essence of various attention methods, and summarized how the attention mechanism is implemented. The huge datasets and evaluation criteria that are regularly utilized in practice are summarized. Even though image captioning can be used for image retrieval, video caption, and video movement, and a wide range of image caption systems are currently available, experimental results suggest that this task still requires higher performance systems and improvement.

CONCLUSION

The CNN-LSTM model was created to automatically generate captions for the input images. This concept can be used in a wide range of situations. We learned about the CNN model, and LSTM models, and how to overcome previous limitations in the field of graphical image captioning by building a CNN-LSTM model capable of scanning and extracting information from any input image and transforming it into a single line sentence in natural language English. The algorithm attention and how the attention mechanism is used were the main topics of discussion. I was able to successfully create a model that is a major improvement above the earlier image caption generator.